

Low-pass Filter and Gain

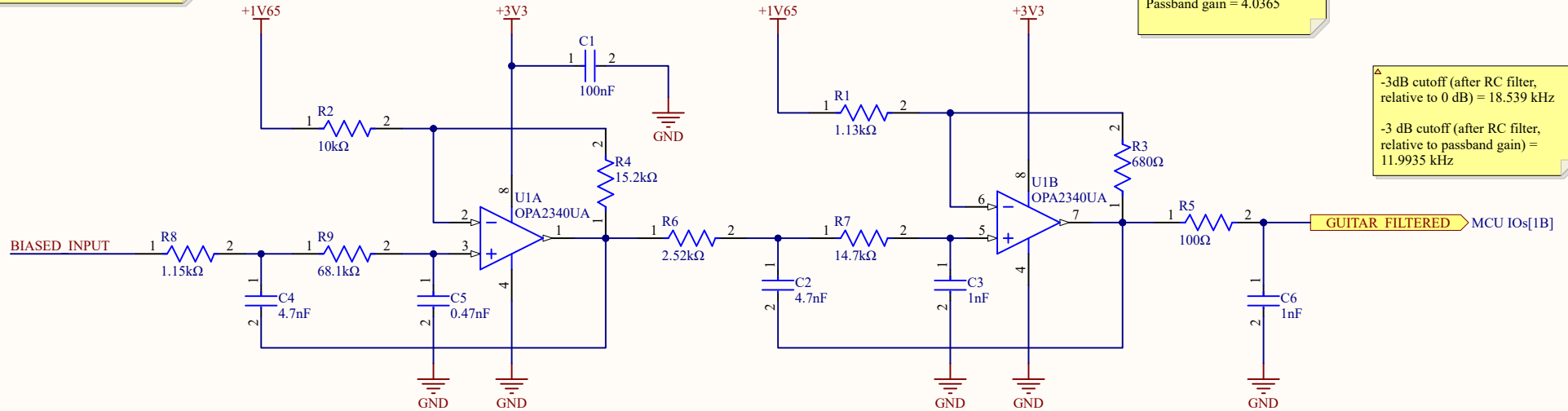
4th-order Butterworth low-pass filter using Sallen-Key topology

First stage $Q = 0.54056$
Target $Q = 0.541196$
% error = 0.12%

Second stage $Q = 1.307384$
Target $Q = 1.30656296$
% error = 0.06%

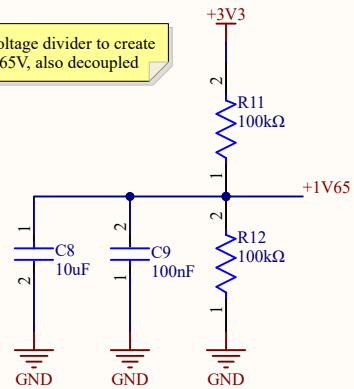
Passband gain = 4.0365

-3dB cutoff (after RC filter, relative to 0 dB) = 18.539 kHz
-3 dB cutoff (after RC filter, relative to passband gain) = 11.9935 kHz



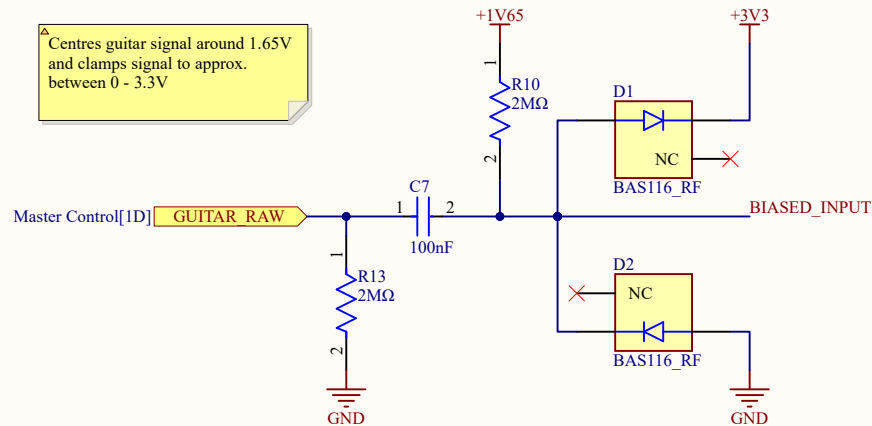
1.65V Net

Voltage divider to create 1.65V, also decoupled



Input Bias and Clamping

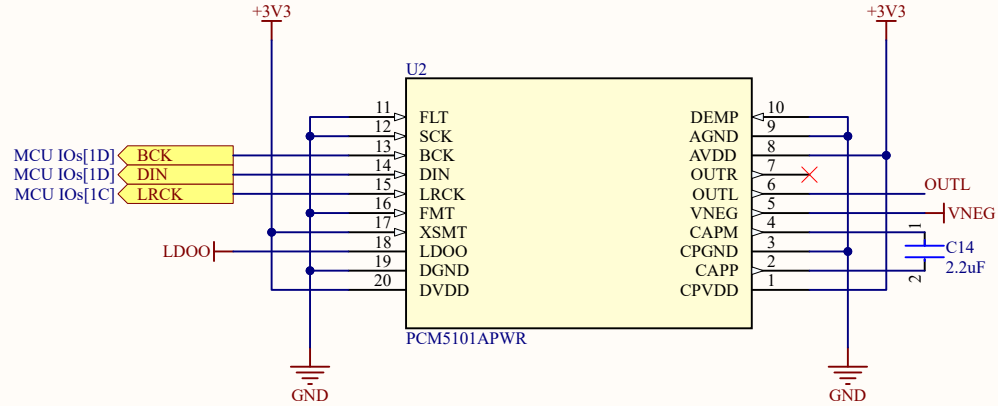
Centres guitar signal around 1.65V and clamps signal to approx. between 0 - 3.3V



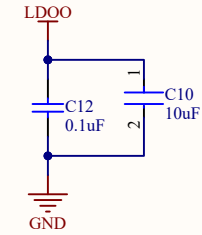
Title		
Analog Input		
Size	Number	Revision
A4	1	3.3
Date:	7/30/2026	Sheet of 6
File:	C:\Users\...\Analog Input.SchDoc	Drawn By: Leo Chinchilla

DAC

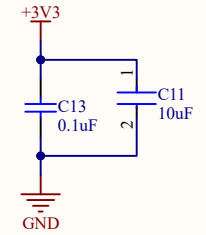
External 32-bit DAC for synthesizer audio



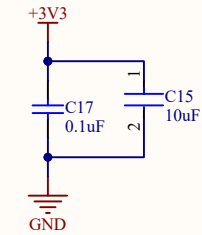
LDOO



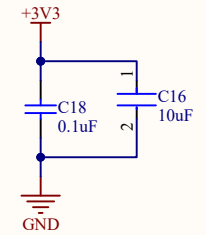
DVDD



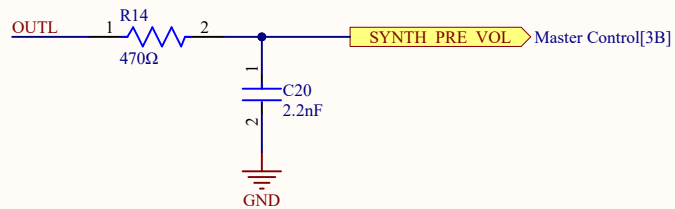
CPVDD



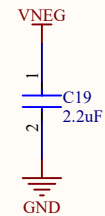
AVDD



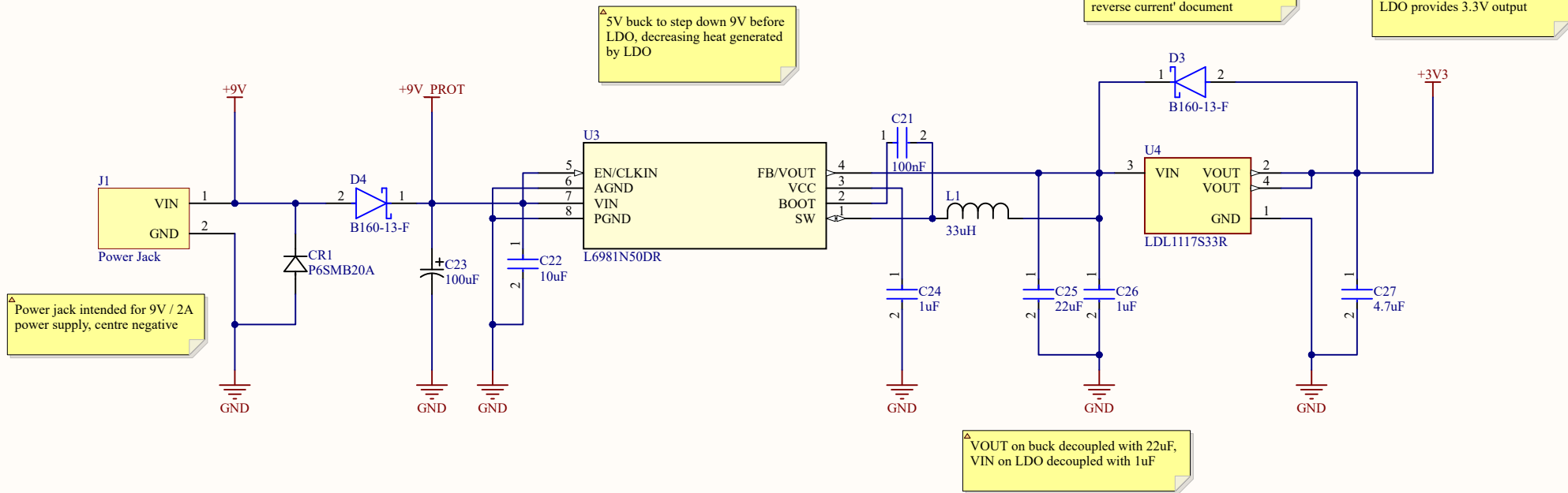
DAC Output



VNEG

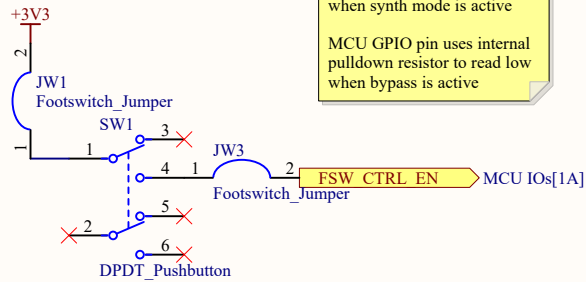


Title Analog Output			
Size A4	Number 2	Revision 3.3	
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Title Board Power			
Size A4	Number 3		Revision 3.3
Date:	7/30/2026		Sheet of 6
File:	C:\Users\...\Board Power.SchDoc		Drawn By: Leo Chinchilla

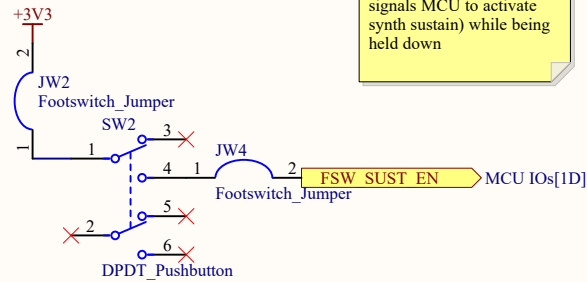
Control Footswitch



△ Latching footswitch used to send logic HIGH to MCU when synth mode is active

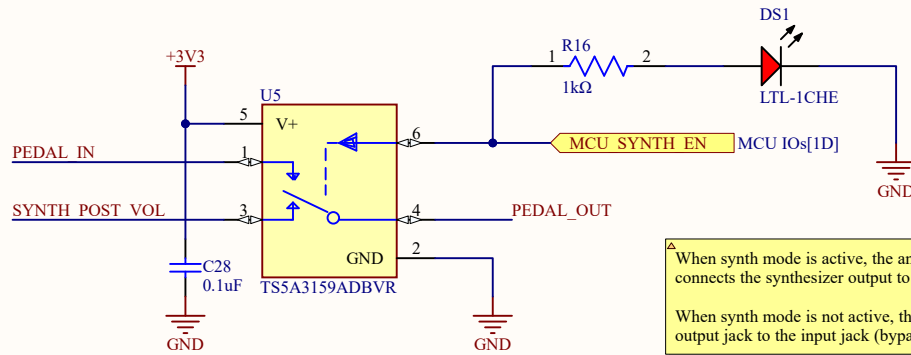
MCU GPIO pin uses internal pulldown resistor to read low when bypass is active

Sustain Footswitch



△ Momentary footswitch that signals MCU to activate synth sustain) while being held down

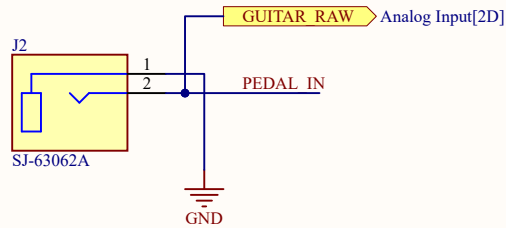
Control Multiplexer



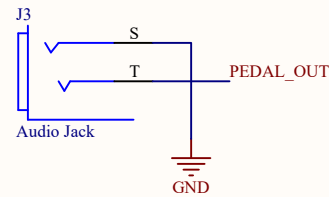
△ When synth mode is active, the analog switch IC connects the synthesizer output to the output jack.

When synth mode is not active, the switch connects the output jack to the input jack (bypass mode)

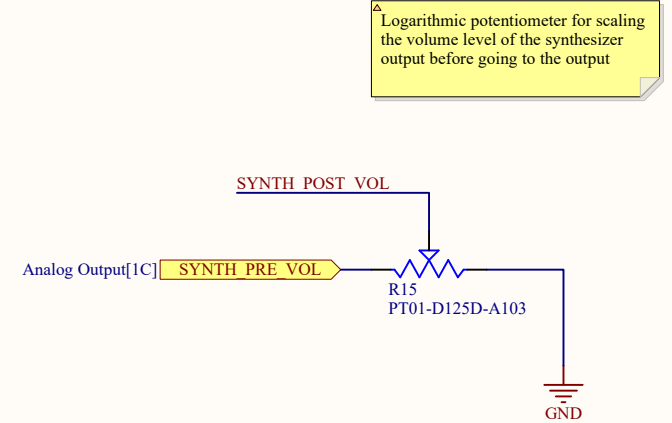
Pedal Input



Pedal Output



Potentiometer



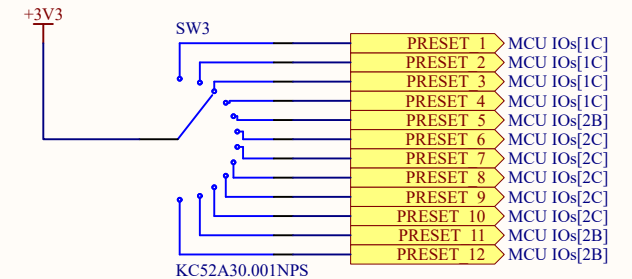
△ Logarithmic potentiometer for scaling the volume level of the synthesizer output before going to the output

Preset Selection

△ Each throw of the rotary switch connects to a GPIO pin

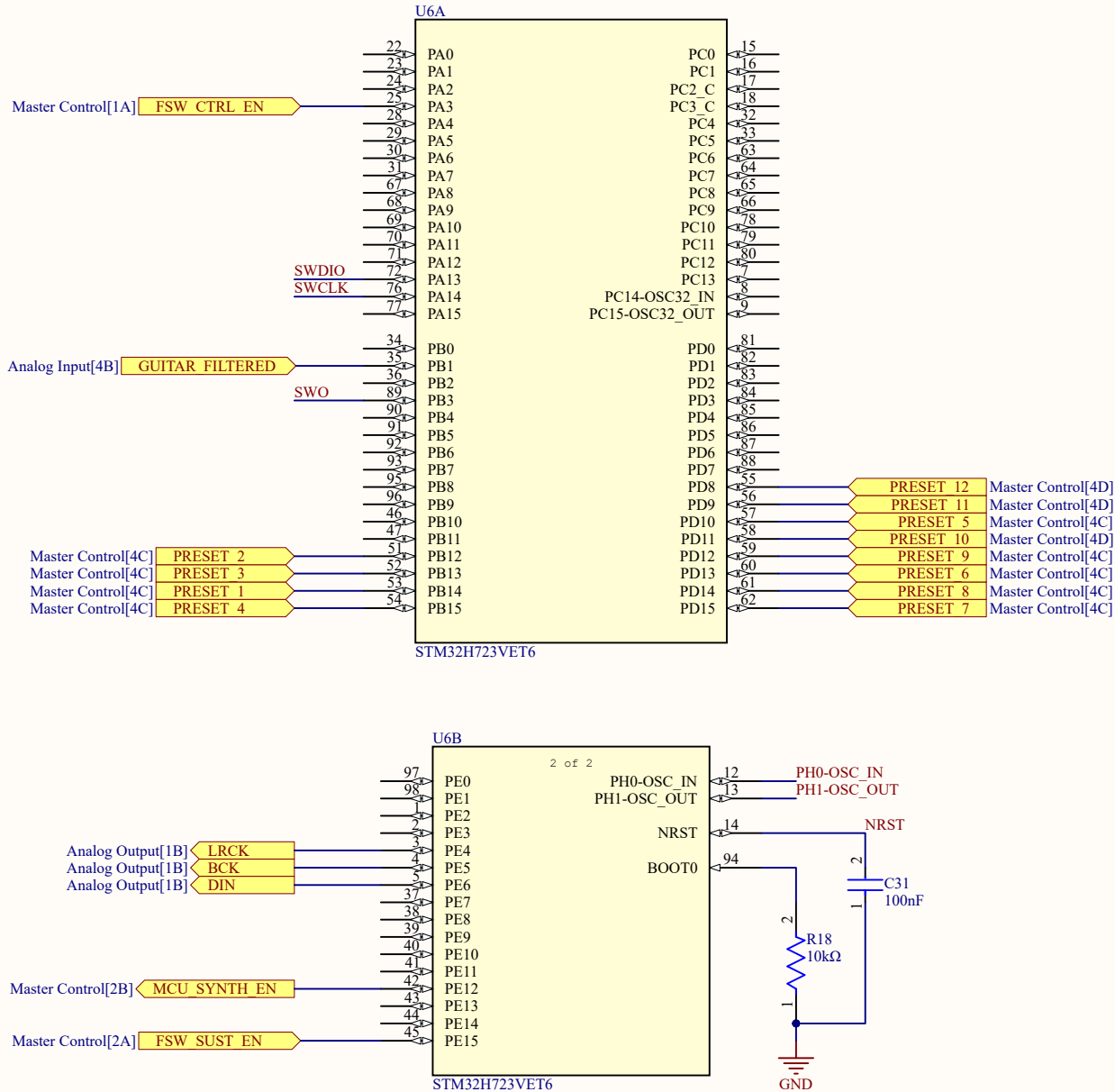
Each GPIO pin is in a pulldown configuration

When the +3.3V pole lands on a throw, that GPIO pin is driven high, and we know that the specific preset is being selected

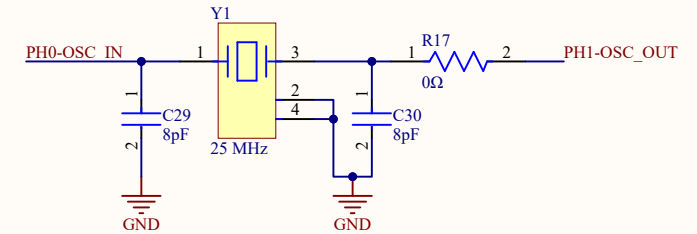


Title		
Master Control		
Size	Number	Revision
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Date:	7/30/2026	Sheet of 6
File:	C:\Users\...\Master Control.SchDoc	Drawn By: Leo Chinchilla

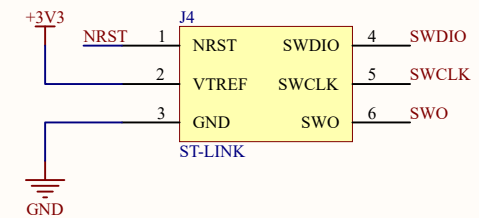
MCU IOs



Crystal Oscillator

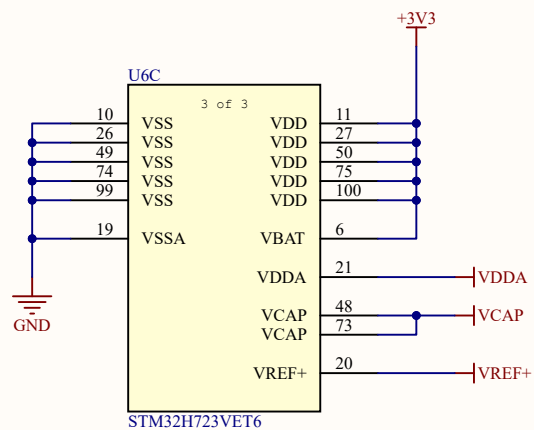


SW Debug



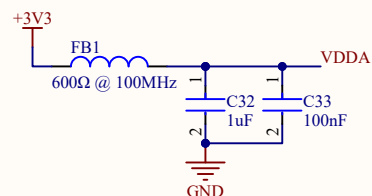
Title			MCU IOs	
Size	Number	Revision		
A4	5	3.3		
Date:	7/30/2026	Sheet of	6	
File:	C:\Users\...\MCU IOs.SchDoc	Drawn By:	Leo Chinchilla	

MCU Power



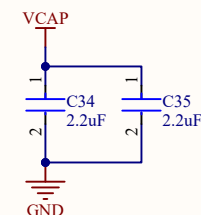
VDDA

△ VDDA decaps as close as possible to the VDDA pin. VDDA connects to VDD through the ferrite bead



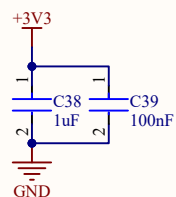
VCAP

△ Placed as close as possible to VCAP pins



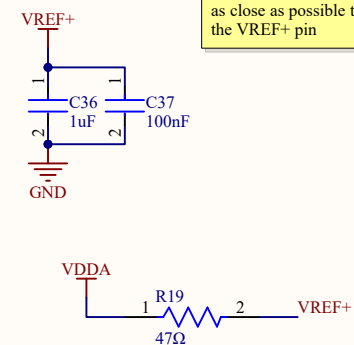
VBAT

△ VBAT decaps to be placed as close as possible to the VBAT pin



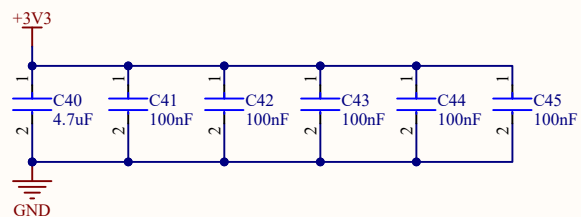
VREF+

△ VREF+ tied to VDDA with a 47ohm resistor, decaps should be placed as close as possible to the VREF+ pin



VDD

△ VDD decaps to be placed as close as possible to VDD pins. 4.7uF decap before any one of the VDD pins, pick most suitable



Title MCU Power			
Size A4	Number 6	Revision 3.3	
Date:	7/30/2026	Sheet of	6
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